

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO3: Construct regular expressions, and use the pumping lemma and closure properties to analyze regular languages. (BL : 3)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 25

Duration: 1 Hour

CASE STUDY

Code of Conduct:

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I 192372044 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

1. Consider a case study in parsing nested structured expressions using a Context-Free Grammar (CFG).
Given the grammar

$S \rightarrow (L) \mid a$

$L \rightarrow L, S \mid S,$

Designing a Grammar for Arithmetic Expressions?

Case study:

A compiler development team is creating a parser for a calculator application that supports addition, subtraction, multiplication and division and parentheses. The initial grammar produces ambiguity for expressions such as $a + b * c$.

Ans: Ambiguous Grammar:

$$E \rightarrow E + E \mid E * E \mid (E) \mid id$$

This grammar is ambiguous because the expression $a + b * c$ can have multiple parse trees.

Optimized CFG:

$$E \rightarrow E + T \mid E - T \mid T$$

$$T \rightarrow T * F \mid T / F \mid F$$

$$F \rightarrow (E) \mid id$$

Optimization performed:

- Separate non-terminals for precedence levels
- T handles multiplication/division
- E handles subtraction/addition
- Removes ambiguity
- Supports efficient topdown (or) bottom-up parsing.

2. Mini Language programming parser development

Case study:

A start up is designing a small scripting language for automating IoT devices. The parser frequently reports syntax conflicts during compilation.

Ans:

Initial Grammar with left Recursion:

$$stmt \rightarrow stmt stmt \mid stmt$$

Optimized CFG: $stmt \rightarrow stmt \text{ } stmt'$
 $stmt' \rightarrow stmt \text{ } stmt' \mid \epsilon$

CFG for language constructions:

Loop $\rightarrow$ while (Condition) { stmt }

Cond $\rightarrow$ if (Condition) { stmt }

Optimization performed:

- Eliminated left recursion
- Reduced shift-reduce conflicts
- Improved predictive parsing.

3) Grammar optimization for SQL Query validation.

Case study: A database company is developing a query validation tool for "simplified SQL commands."

Ans:

Initial Grammar:

$Q \rightarrow \text{SELECT } A \text{ FROM } T \text{ WHERE } C$

Optimized CFG:

$Q \rightarrow \text{SELECT Fields FROM Table Optwhere}$

$\text{optwhere} \rightarrow \text{WHERE Condition} \mid \epsilon$

Optimization performed:

- Introduced optional productions.
- Reduced Redundancy
- Simplified parse design
- Improved syntax validation speed

Example Accepted Query:

SELECT Name FROM student WHERE mark > 50

4) Natural Language processing chatbot.

Case study:

A chatbot processes simple English sentences like

"Students learn ToC!"

CFG Design:

$S \rightarrow NP VP$

$NP \rightarrow Noun$

$VP \rightarrow Verb NP$

$Noun \rightarrow \text{students} [ToC]$

$verb \rightarrow \text{learn}$

optimization performed:

- Restricted sentence order
- Invalid patterns are rejected
- Improved grammatical correctness.

valid sentence:

Students Learn ToC

Invalid sentence:

Learn Students ToC.

5. parse & design for HTML-like Markup Language.

Case study: A web editor company is building a parser for a simplified Markup Language with Nested tags.

CFG For Nested Tags:

$S \rightarrow \langle \text{tag} \rangle S \langle / \text{tag} \rangle | \text{text} | \epsilon$

Example:

$\langle \text{tag} \rangle$

$\langle \text{tag} \rangle \text{text} \langle / \text{tag} \rangle$

$\langle / \text{tag} \rangle$

optimization performed:

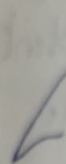
- Recursive production supports nesting
- Ensures proper opening and closing tags

- Detects Invalid nesting structures.

Invalid Example:

<tag> <|> <|> <|> </tag> </br>

Benefit:



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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandabl e with minor issues	Somewhat unclear or poorly structured	Difficult to understand

92/100

